What makes motion hard to stop?

antocciphysics

UNIT 3 / DAY 1 / HONORS

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare momentum changes in a controlled stopping situation.

SUCCESS CRITERIA - I can...

1. Compare mass and velocity while controlling other variables.
2. Use momentum change to compare stopping force for equal times.
3. Avoid claiming that momentum alone determines damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

Two ideal Sun-orbits have $a_B=3\ a_A$. Find T_B/T_A. Explain why the same normalized Sun-orbit equation is not automatically valid around another body.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $p = mv$; $F_{avg} = \Delta p/\Delta t$.
Momentum p includes direction. For a fixed stopping time, greater momentum change needs greater average net force.

Worked reasoning: A 2 kg cart and a 4 kg cart move right at 3 m/s. Their momenta mv are 6 and 12 kg·m/s. Stopping the second requires twice the momentum change.

Your turn: Draw the velocity and momentum arrows for both carts. Explain which pair has equal lengths.

Marzano check - circle: ① ② ③

03 Represent and interpret

Carts A and B both move right at 2 m/s. A has mass 1 kg; B has mass 3 kg. Which has more momentum? State the controlled quantity.



A: 1 kg; 2 m/s B: 3 kg; 2 m/s

Marzano check - circle: ① ② ③

04 Solve and justify

Compare A:(1 kg,+2 m/s), B:(3 kg,+2 m/s), C:(1 kg,+4 m/s). Each stops in .50 s. Find signed changes in momentum and average forces.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

Two objects have equal momentum but strike different materials over different times. Can equal momentum alone establish equal damage? Explain.

Marzano check - circle: 1 2 3

06 Choose a method

Two equal 4000 kg masses are 6 m apart. Use $G=6.67\times10^{-11}$ N·m²/kg². Find force. By what factor is an answer wrong if r rather than r² is used?

Marzano check - circle: 1 2 3

07 Exit check

A 3 kg cart moving right at 4 m/s stops in 4 s. Find signed impulse and average force with right positive.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 0.5 kg cart at +8 m/s and a 2 kg cart at +2 m/s each stop in 0.2 s. Compare their momentum changes and average stopping forces.

Marzano check - circle: 1 2 3